

South Australian Electricity Market Performance Overview

Revenue, generation, price exposure and technology performance | July 2022 to June 2024

\$411.57M

Total Revenue

2.38M

Total Generation MWh

\$100.88

Time Weighted Avg Price

\$128.79

Market VWP

48.23K

Negative Price Intervals

Fiscal Year

All

Technology

All

Date

7/1/2022

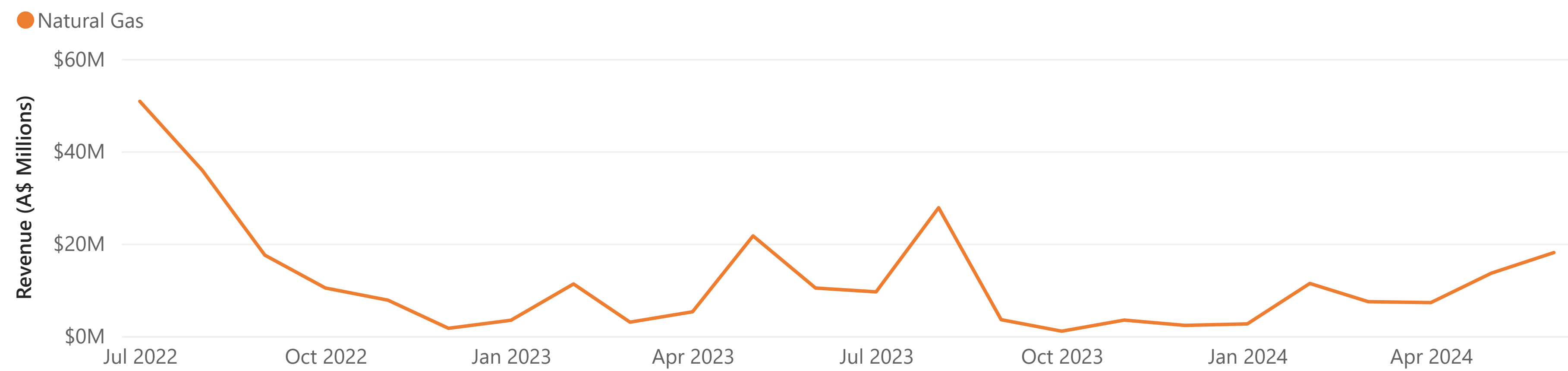


6/30/2024



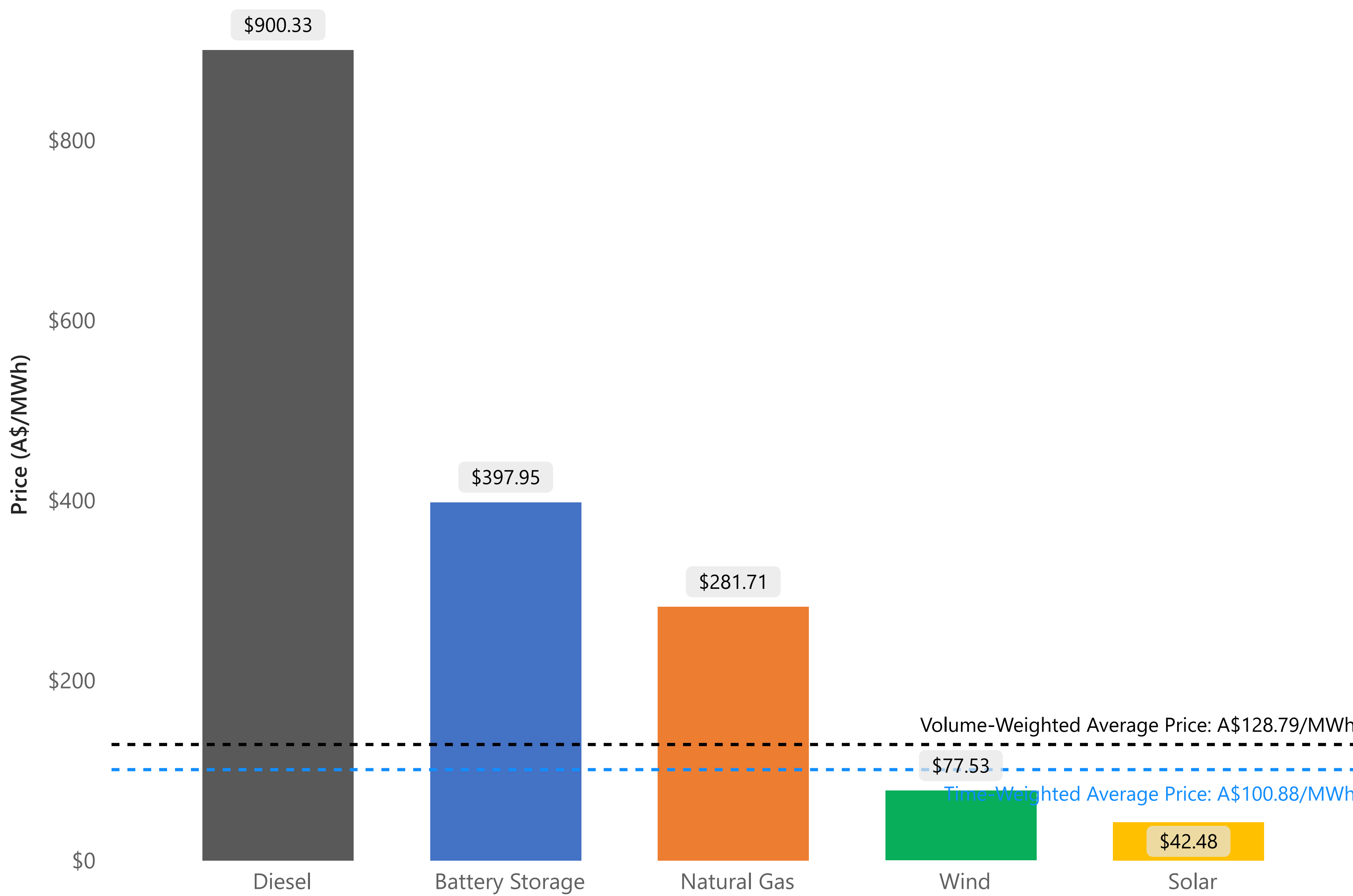
Monthly Revenue Trends by Generation Technology

Monthly wholesale market revenue variability across technologies from July 2022 to June 2024



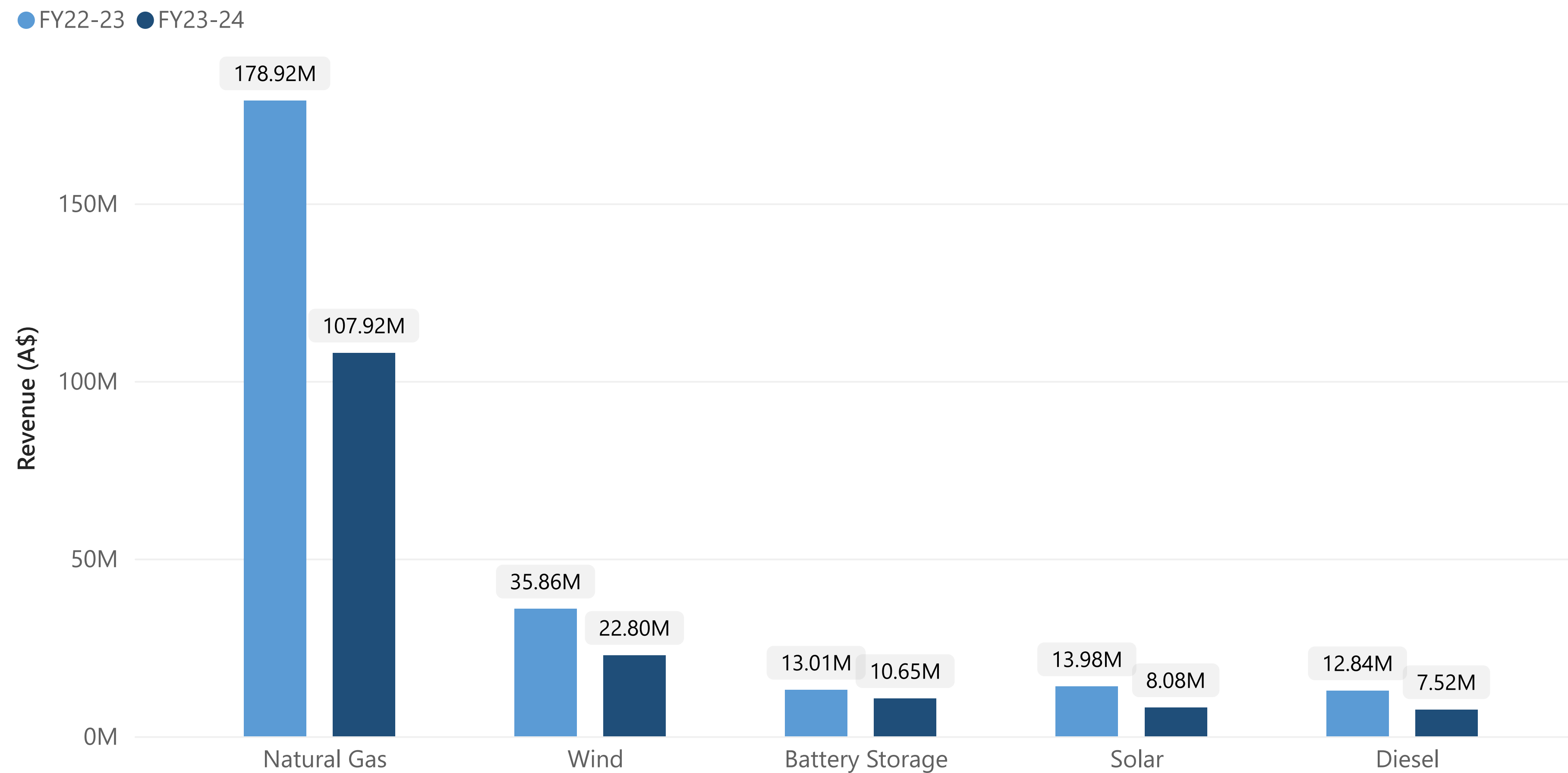
Dispatch-Weighted vs Market Benchmark Prices by Technology

Technology-level capture prices compared with market benchmark prices



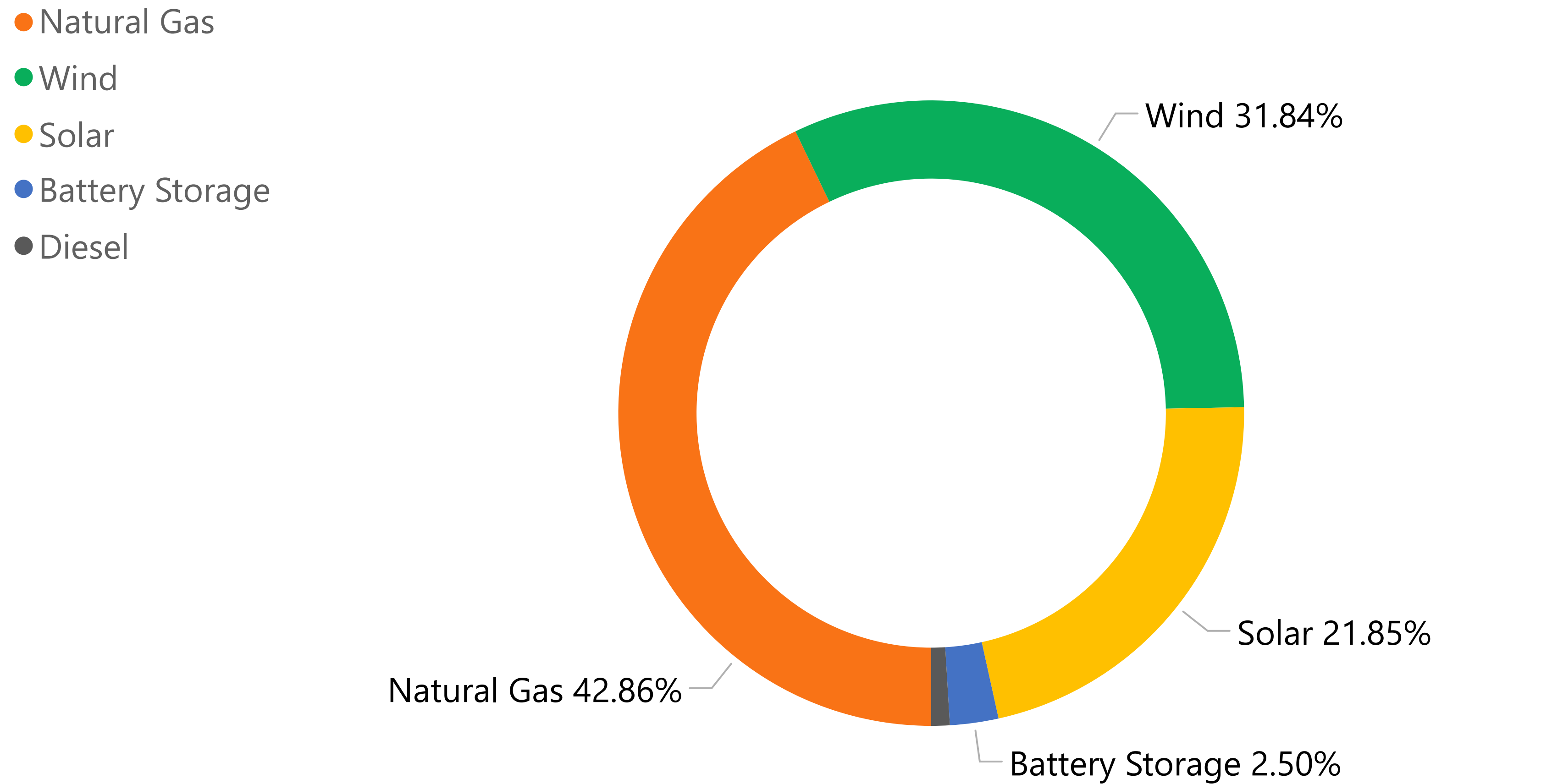
Annual Revenue by Generation Technology

Comparison of total wholesale market revenue across FY2022–23 and FY2023–24



Generation Mix by Technology

Technology contribution to total generation volume



Technology Performance, Risk and Utilisation Analysis

Comparison of revenue stability, utilisation, capture price and negative price exposure | July 2022 to June 2024

Fiscal Year

All

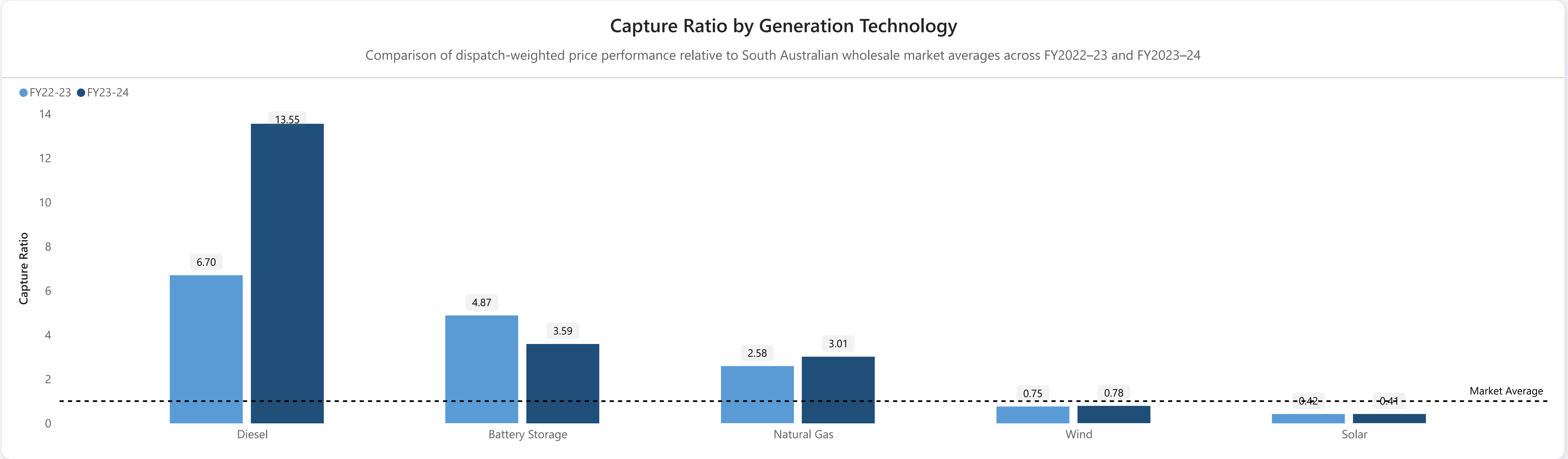
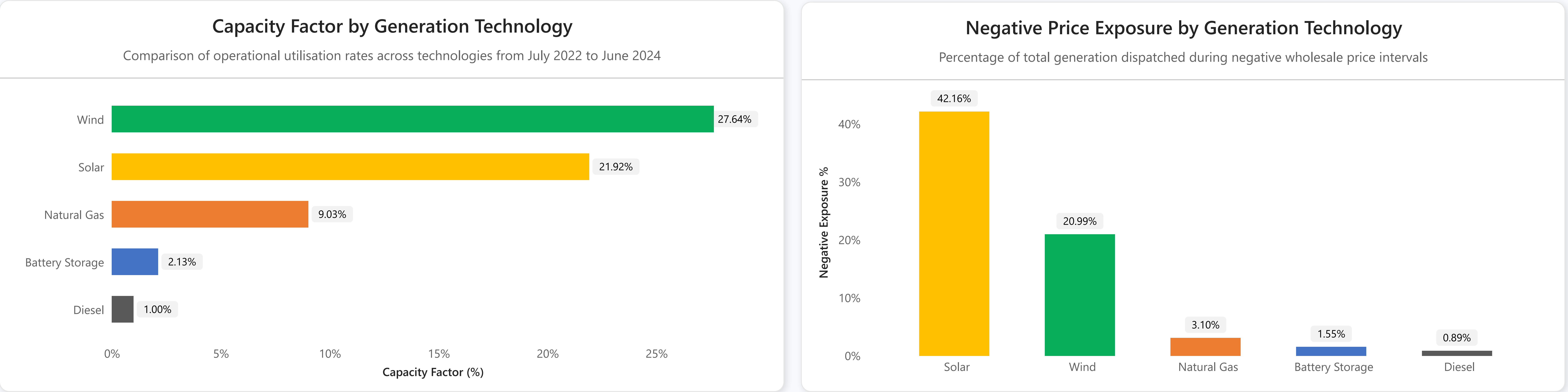
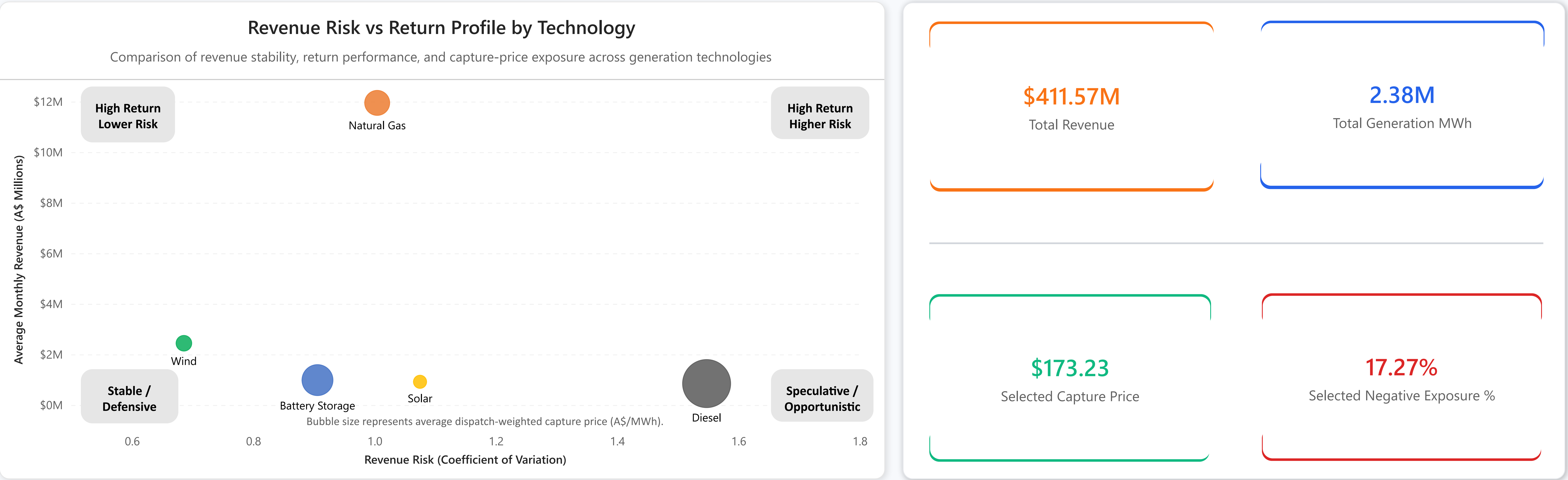
Technology

All

Date

7/1/2022

6/30/2024



Dispatch Timing and Emissions Profile

Hourly dispatch behaviour, price alignment and estimated carbon emissions | July 2022 to June 2024

Fiscal Year

All

Technology

All

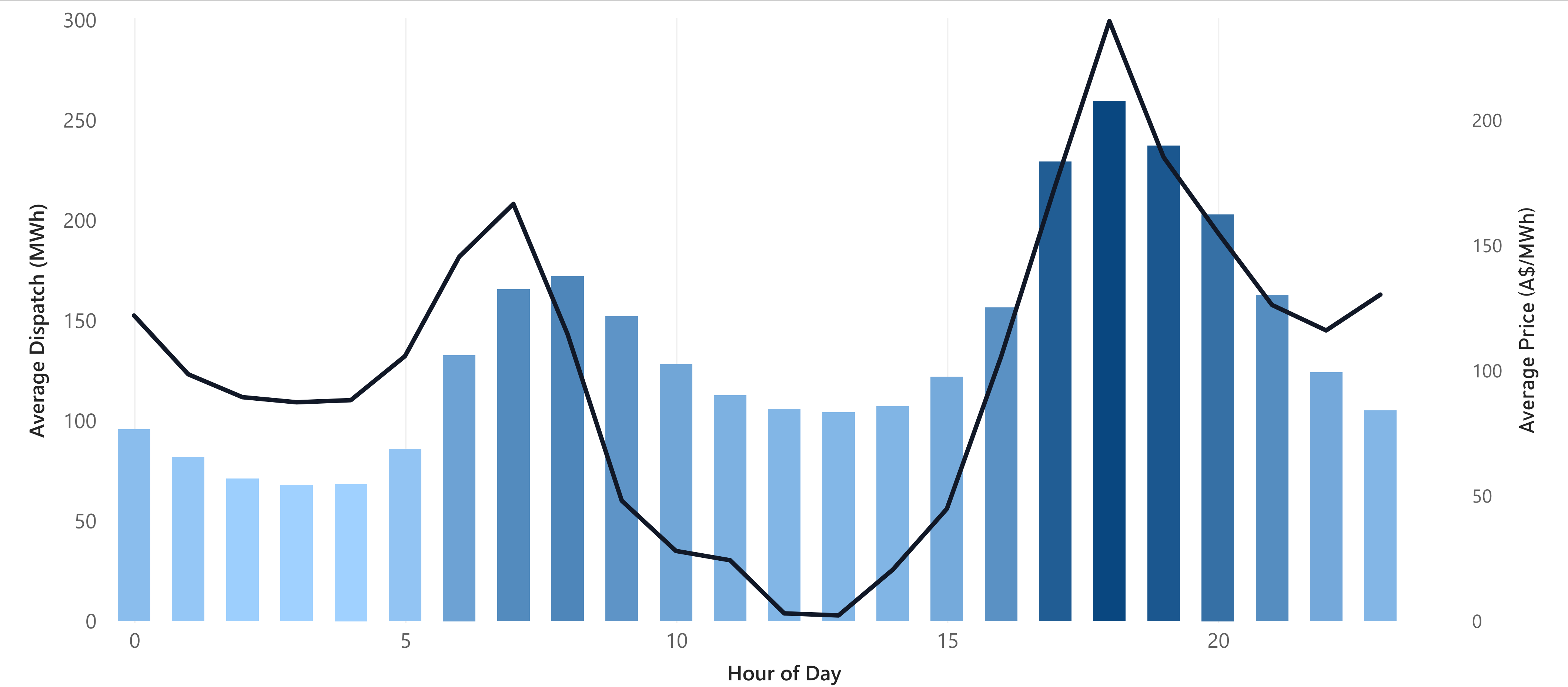
Date

7/1/2022

6/30/2024

Hourly Dispatch Timing vs Market Price

Dispatch timing explains differences in technology capture prices and market exposure



\$411.57M

Total Revenue

2.38M

Total Generation MWh

\$173.23

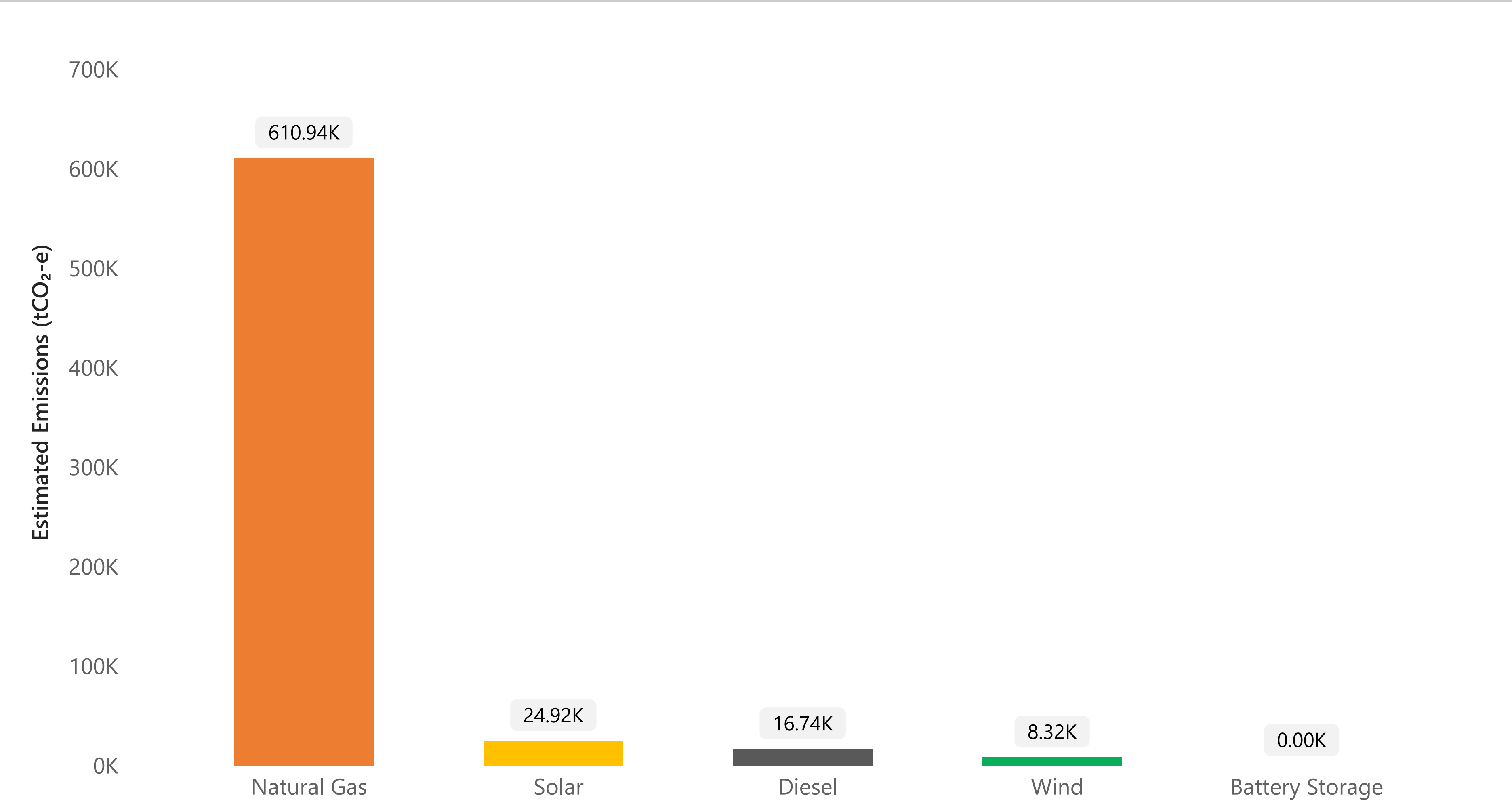
Capture Price

660.91K

Emissions (tCO₂-e)

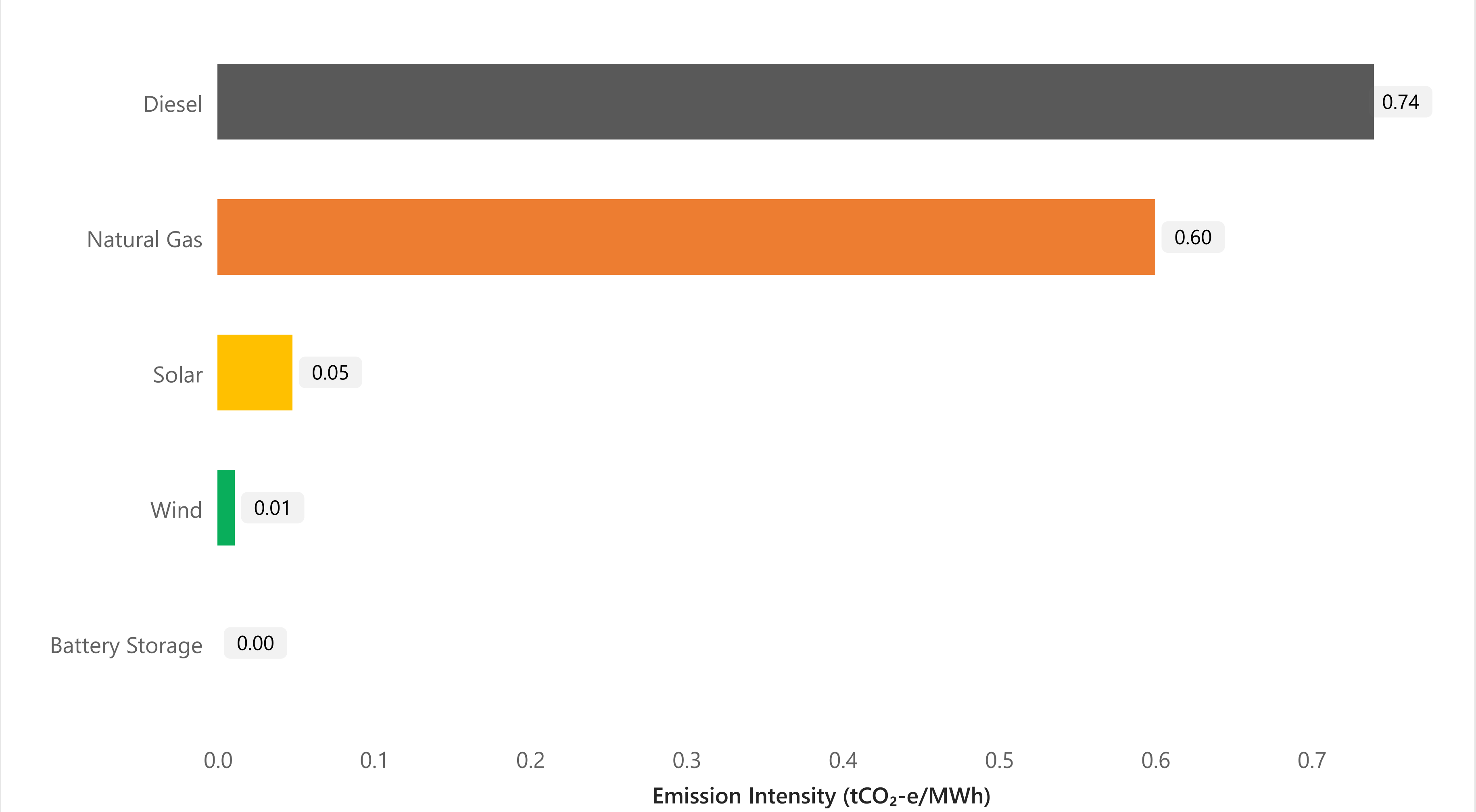
Estimated Total Carbon Emissions by Generation Technology

Approximate operational emissions generated across South Australian generation technologies from July 2022 to June 2024



Carbon Emission Intensity by Generation Technology

Comparison of estimated operational carbon emissions across South Australian generation technologies



Key Findings and Strategic Insights

Summary of market performance, technology risk, price exposure and emissions implications

Insights

- 1) Revenue concentrated in dispatchable generation:** Natural gas generated the highest total wholesale revenue across the analysis period, even though wind and solar contributed substantial generation volume. This shows that revenue performance was not driven by output alone. Dispatch timing and exposure to high-price intervals were more important in determining market value.
- 2) Diesel and battery captured premium prices:** Diesel and battery storage achieved the highest dispatch-weighted capture prices because their dispatch was concentrated during higher-price periods. However, diesel remained highly opportunistic because its total generation volume was low and revenue volatility was high. Battery storage showed stronger strategic value due to its ability to align dispatch with evening and peak-price periods.
- 3) Solar and wind faced price cannibalisation risk:** Solar and wind earned lower capture prices because a large share of their output occurred during lower-price or negative-price intervals. Solar was most exposed to negative prices, reflecting midday generation concentration and oversupply risk. This creates a revenue challenge for renewable-heavy portfolios without storage or demand-side flexibility.
- 4) Utilisation and revenue did not move together:** Natural gas produced the largest estimated total emissions because it combined high dispatch volume with a relatively high emissions intensity. Diesel had the highest emissions intensity per MWh, but its low generation volume limited its total emissions contribution compared with gas.
- 5) Emissions were dominated by natural gas:** Natural gas produced the largest estimated total emissions because it combined high dispatch volume with a relatively high emissions intensity. Diesel had the highest emissions intensity per MWh, but its low generation volume limited its total emissions contribution compared with gas.

Recommendations

- 1)** Battery storage should be prioritised as a strategic flexibility asset because it can shift energy toward higher-price periods. This can improve capture price performance, reduce exposure to low-price intervals and support better integration of renewable generation.
- 2)** Solar-heavy portfolios should actively manage negative price exposure through storage, curtailment strategies, contracting, demand-side flexibility or load-shifting opportunities. This is important because high generation volume alone may not protect revenue when prices are low or negative.
- 3)** Natural gas delivered strong revenue performance, but it also created the largest estimated emissions impact. Future investment and dispatch decisions should consider both financial returns and carbon exposure, especially as market and policy pressure around emissions increases.
- 4)** Technology performance should be assessed using capture price, capture ratio, negative price exposure and revenue volatility, not only total generation. These indicators provide a better view of market value and risk across different generation technologies.

The analysis shows that South Australian electricity market performance was shaped by dispatch timing, price exposure, flexibility and emissions intensity. Dispatchable technologies such as diesel, battery storage and natural gas captured higher prices because they operated during more valuable market intervals. In contrast, wind and solar contributed significant generation volume but faced lower capture prices and higher negative price exposure, especially solar during midday periods.

Overall, the strongest strategic opportunity is to improve flexibility in the system. Battery storage and demand-side flexibility can help shift renewable energy toward higher-price periods, reduce negative price exposure and improve market value. At the same time, the emissions results show that high-revenue technologies can also carry high carbon impact, particularly natural gas. Therefore, future market strategy should balance revenue optimisation, technology risk, renewable integration and emissions reduction rather than relying on generation volume alone.